CLINICAL · APPROACH TO

Approach to tinnitus

Tinnitus is a sound with no outside source, and it is a symptom rather than a disease. Two kinds need imaging: tinnitus in one ear, and tinnitus that pulses with the heartbeat, because a tumour or a vascular lesion may lie behind them. Most other tinnitus needs good hearing care, explanation and sound therapy, not drugs.

AT A GLANCE

- Find the cause first.
- Pulsatile or one-sided tinnitus needs imaging.
- Hearing aids, sound therapy and counselling help most.
- Ask about mood.

1 CHAPTER ONE Overview

A sound no one else can hear.

1.1 Definition

The perception of sound, ringing, hissing, buzzing or roaring, in the absence of an external source.

1.2 Epidemiology

Between 10 and 15% of adults have tinnitus, but only about one in thirty find it at least moderately annoying and about one in two hundred are severely affected. Most have some hearing loss, often age-related or noise-induced. About half of those with troublesome tinnitus also have reduced tolerance of sound.

1.3 Why it matters

It can be the first symptom of a vestibular schwannoma, a glomus tumour or a dural arteriovenous fistula. And severe tinnitus disrupts sleep, concentration and mood.

2 CHAPTER TWO Classification

Who can hear it?

2.1 Subjective tinnitus

Heard only by the patient: by far the commonest.

2.2 Objective tinnitus

Can be heard by the examiner with a stethoscope or a Toynbee tube.

- **Vascular:** pulsatile, from arterial or venous flow.
- **Muscular:** clicking from palatal myoclonus, or from spasm of the stapedius or tensor tympani.
- **Patulous Eustachian tube:** a blowing sound with breathing.
- **Temporomandibular joint:** clicks with jaw movement.

2.3 A normal phenomenon

Sudden brief unilateral tapering tinnitus, a few seconds of ringing in one ear with fullness, happens to most people and needs no investigation.

3 CHAPTER THREE Causes

Ear, body and blood vessels.

3.1 Otological

Wax, middle ear fluid, otitis media, otosclerosis, Ménière's disease, presbycusis, noise-induced hearing loss, sudden hearing loss, ototoxicity and vestibular schwannoma.

3.2 General

Anaemia, hypertension, thyroid disease, hyperlipidaemia, vitamin B12 deficiency, head injury and multiple sclerosis.

3.3 Drugs

Aspirin and other salicylates, quinine, aminoglycosides, loop diuretics and some chemotherapy. Most drug-induced tinnitus reverses when the drug is stopped.

3.4 Pulsatile tinnitus

CAUSES OF PULSATILE TINNITUS

- ARTERIAL **Carotid stenosis, dissection, aneurysm** aberrant internal carotid artery, persistent stapedial artery
- VENOUS **High or dehiscent jugular bulb** sigmoid sinus diverticulum or dehiscence, idiopathic intracranial hypertension
- TUMOUR **Glomus tympanicum or jugulare** other vascular tumours
- SHUNT **Dural arteriovenous fistula** arteriovenous malformation
- HIGH OUTPUT **Anaemia, thyrotoxicosis, pregnancy** conductive hearing loss makes body sounds louder

3.5 How it arises

Reduced input from a damaged cochlea leads to reorganisation and increased spontaneous activity in the central auditory pathways. Everyone has some random activity in these pathways: in a silent room most people soon hear a faint hum or ringing. Normally, subconscious filters ignore it. When the brain labels the sound as a threat, the limbic and autonomic systems respond, attention is drawn to it, and a vicious circle of awareness, anxiety and amplification follows. This model explains why the loudness of tinnitus predicts distress so poorly, and why treatment aims at the reaction rather than the sound.

3.6 Disorders of sound tolerance

Hyperacusis is intolerance of everyday sounds that others find comfortable, often with tinnitus. **Misophonia** is a strong emotional reaction to specific sounds, often mouth sounds such as chewing, regardless of loudness. **Phonophobia** is fear of particular sounds, and **recruitment** is the abnormally rapid growth of loudness in a cochlear hearing loss. Facial palsy, by removing the stapedial reflex, can cause hyperacusis. Sound tolerance is treated with explanation, graded sound exposure and cognitive behavioural therapy; ear plugs worn constantly make it worse.

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4 CHAPTER FOUR Assessment

Character, side, and mood.

4.1 History

Is it in one ear or both? Is it pulsatile? Is there hearing loss, vertigo or fullness? Noise exposure and drugs? How much does it disturb sleep, work and mood?

> **ASK ABOUT MOOD**
> Severe tinnitus is linked with anxiety and depression. Ask directly how the patient is coping, including any thoughts of self-harm, and arrange support where needed.

4.2 Examination

Otoscopy under the microscope (a red mass behind the drum, a blue drum, wax), tuning forks, cranial nerves, and auscultation over the ear, orbit and neck. Pressing on the carotid stops arterial tinnitus; pressing on the jugular vein or turning the head stops a venous hum. A rhythm with breathing points to a patulous tube; clicks with palatal movement mean myoclonus.

4.3 Investigations

4.3.1 Audiometry

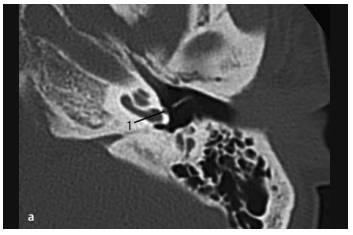
Pure-tone and speech audiometry, tympanometry, and tinnitus matching for pitch and loudness. Loudness and pitch do not predict how distressing it is.

4.3.2 Imaging

- **One-sided tinnitus or asymmetric hearing:** MRI of the internal auditory canals.
- **Pulsatile tinnitus:** CT angiography, MR angiography and venography; formal angiography if a dural fistula is suspected.
- **Palatal myoclonus:** MRI of the posterior fossa.

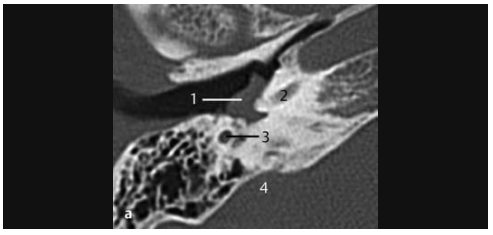
IMAGING IN THE APP 3 cases

A persistent stapedial artery, glomus tympanicum and vestibular schwannoma.



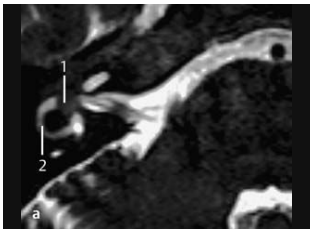
Persistent Stapedial Artery

Variants · 4 images



Glomus Tympanicum (Paraganglioma)

Tumours · 11 images



Vestibular Schwannoma

Tumours · 13 images

4.3.3 Blood tests

As suggested by the history: full blood count, thyroid function, lipids, glucose, B12.

4.3.4 Long-duration tympanometry

Shows rhythmic compliance changes in myoclonus or a patulous tube.

5 CHAPTER FIVE Management

Treat the cause, then the reaction.

5.1 Treat the cause

Remove wax, treat otitis media, operate on otosclerosis, stop ototoxic drugs where possible, and treat anaemia or thyroid disease.

5.2 Hearing aids

The first step whenever there is hearing loss: restoring external sound reduces the contrast with the tinnitus.

OPERATIVE GUIDES IN THEATRE 3 operations

Bone-Anchored Hearing Devices
Bone-anchored hearing
Implantable Middle Ear Devices
Implantable devices
Cochlear Implantation in Adults
Cochlear implant

5.3 Explanation and counselling

Explaining that tinnitus is common, real and not a sign of a brain tumour (once excluded) relieves much of the distress.

5.4 Sound therapy

5.4.1 Masking

Background sound, a bedside sound generator or a wearable masker, alone or combined with a hearing aid.

5.4.2 Tinnitus retraining therapy

Directive counselling plus low-level broadband sound, aiming at habituation: the tinnitus is still there but no longer noticed or disturbing.

WHERE THE BOOKS DIFFER · RETRAINING SOUND EXPOSURE

Dhingra	Lee
At least 8 hours a day for 18 to 24 months.	Up to 16 hours a day for about a year.

5.5 Psychological therapy

Cognitive behavioural therapy has the strongest evidence for reducing distress; relaxation and biofeedback also help. Treat anxiety, depression and insomnia.

5.6 Specific objective causes

- **Dural arteriovenous fistula:** embolisation, surgery or radiosurgery; fistulas with cortical venous drainage can bleed.
- **Palatal myoclonus:** muscle relaxants or botulinum toxin.
- **Stapedial or tensor spasm:** muscle relaxants, or division of the tendon.
- **Patulous tube:** see the Eustachian tube topic.

5.7 Treatments without proven benefit

No drug reliably suppresses tinnitus, and betahistine is not recommended for it. Transcranial magnetic stimulation and implanted stimulators remain investigational. A cochlear implant often reduces tinnitus in patients who qualify for it because of their hearing loss.

5.8 Stepped care

Severity is measured with a validated questionnaire such as the **Tinnitus Handicap Inventory** or the Tinnitus Functional Index. Most patients need only explanation, hearing care and self-help; those with greater distress move to audiologist-led tinnitus support and sound therapy, and then to cognitive behavioural therapy delivered by a psychologist, with access to mental health services for those at risk.

6 CHAPTER SIX Pathway

From symptom to plan.

TINNITUS

1 Pulsatile?

→ **yes, audible to examiner or not**
otoscopy, auscultation, CT angiography or MR angiography and venography

→ **young woman with obesity and headache**
fundoscopy — idiopathic intracranial hypertension

→ **red mass behind the drum**
glomus — imaging

2 Unilateral or asymmetric hearing?

→ **yes**
MRI of the internal auditory meatus

→ **sudden hearing loss with it**
treat as sudden sensorineural loss

3 Bilateral, non-pulsatile

→ audiogram

→ treat wax, effusion, otosclerosis

→ hearing aids for hearing loss

4 **Distress**

- → explanation and self-help
- → sound therapy and tinnitus counselling
- → cognitive behavioural therapy
- → urgent help if risk of self-harm

7 CHAPTER SEVEN Prognosis

Most get better with time.

About a quarter of patients find their tinnitus nearly resolves, half improve, and a quarter stay the same. Habituation is the realistic goal.

PEARL

Most people adapt to tinnitus. What a clinician says in the first consultation strongly shapes how well they do.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Subjective versus objective** tinnitus and examples of each.
- **Causes of pulsatile tinnitus,** and how to investigate them.
- **Unilateral tinnitus:** MRI for schwannoma.
- **Tinnitus retraining therapy** and the aim of habituation.
- **Hyperacusis and misophonia.**

- **Stepped care** and cognitive behavioural therapy.

8.2 Common pitfalls

COMMON PITFALLS

- Telling a patient that nothing can be done.
- Missing pulsatile tinnitus because nobody asked.
- Forgetting to image one-sided tinnitus.
- Ignoring the patient's mood.

Written for this app from Oxford Textbook of Otolaryngology ch. 6, Oxford Specialist Handbook ch. 3, 18, Oxford Handbook of ENT ch. 3, 7, Dhingra ch. 5, 22 and K. J. Lee ch. 18. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

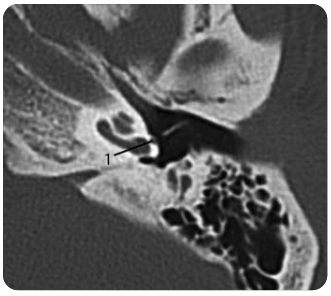
On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · VARIANTS

Persistent Stapedial Artery

- An embryonic branch of the ICA (from the hyoid artery) that persists, crossing the promontory and passing through the obturator foramen of the stapes.
- CT: a small canal over the promontory, an enlarged anterior tympanic segment of the facial canal, and an absent foramen spinosum (the middle meningeal artery arises from the stapedial artery instead).
- Usually found at stapes surgery; leave it alone, as coagulating it risks bleeding and facial or hearing damage.



CT, axial. A retrospective close examination of the CT scan revealed the stapedial artery passing along the wall of the promontory (1), between…

MIDDLE EAR & MASTOID · TUMOURS

Glomus Tympanicum (Paraganglioma)

IMAGING FINDINGS

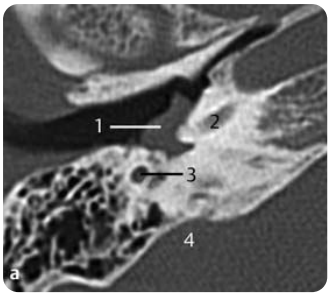
- Small enhancing mass on the cochlear promontory
- Floor of the middle ear intact
- Flow voids only in larger tumours

TOP DIFFERENTIALS

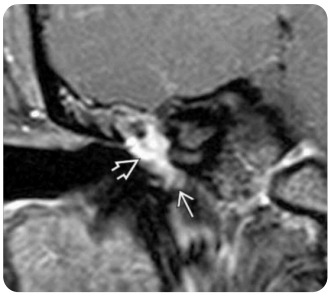
- Glomus jugulare (floor eroded, from the jugular foramen)
- Aberrant internal carotid artery
- Dehiscent jugular bulb

CHECKLIST

- Check the hypotympanic floor and jugular bulb
- Ossicles and extension into the mastoid



CT, axial. On this axial slice at the level of the hypotympanum, a smooth-bordered lesion (1) is seen, without any destruction of the cochlea (2) or…



Coronal T1 C+ FS MR demonstrates a large GTP filling the middle ear cavity. The floor is intact, separating the tumor from the jugular bulb below.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1064.

INNER EAR & IAC · TUMOURS

Vestibular Schwannoma

IMAGING FINDINGS

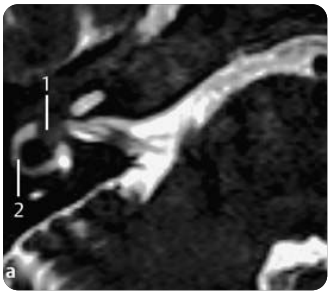
- Enhancing mass centred on the porus, "ice cream on a cone"
- Dark filling defect in bright CSF on high-resolution T2; fundal cap of fluid
- Acute angle with the petrous bone; widened porus; cystic change in larger tumours

TOP DIFFERENTIALS

- Meningioma (broad dural base, dural tail)
- Epidermoid cyst (DWI bright)
- Facial nerve schwannoma (labyrinthine segment)

CHECKLIST

- Size in canal and cistern; brainstem and fourth ventricle (Koos)
- Fundal cap and cochlear nerve canal for hearing preservation



MRI, T2-weighted, axial.A partial lack of fluid in the vestibule and horizontal semicircular canals is evident (1). A mass lesion is present in the vestibul…



Axial T2WI MR reveals a small intracanalicular vestibular schwannoma visualized as a soft tissue intensity mass surrounded by…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222.